1. Analysis of Frames

We have already discussed in the class, the difference between trusses and frames. Though both are assembly of rigid bars, we assume that in trusses the bars are hinged at their ends and loads are acting only at their joints. These assumptions make the bars in trusses two-force members and for equilibrium of two-force members these forces must be equal, opposite, and collinear. Hence, we can always fix the lines of action of forces in truss members along the axis of the bars. The bars will be always in tension or compression only (No bending).

If the rigid bars assembly does not satisfy the above assumptions, we call it as a frame. Since the frame contains one or more members where more than two forces act, we can not consider the line of action of forces in the members as acting along their axes.Therefore, we will not be able to consider the equilibrium of joints as we did in the case of trusses. In frames, we need to consider the free body diagrams of each member, and apply the equilibrium equations, to evaluate the forces acting on each joint. So this method is known as the method of members.

Q.1 Consider the frame shown in figure 1 formed by three bars. BD and AE are pinned at C and load P is acting at the centre of bar DE. The frame rests on a smooth horizontal surface. Calulate the forces in the pins at C,D, and E. Take the distance AB and AD as 1m each.

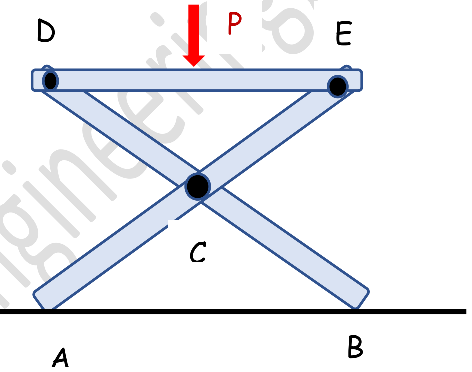


Fig.1

A.1 The bar AE will be experiecing forces at support A and hinged connections at C and E. The member BD will be having forces at support B and hinges at C and D. The bar DE has forces at hinges D and E and the external load P at its centre. So, all the bars are having three forces each.

Let us draw the free body diagrams of each of these bars. Since the support is smooth, there will be only normal reactions R_A and R_B. Each of the hinged joints are assumed to have two components of reactions. The directions of these forces are taken arbitrarilyin one member. But please note that the same forces reverses their directions in the other member joined at that point. This is highlighted by the green dashed line.

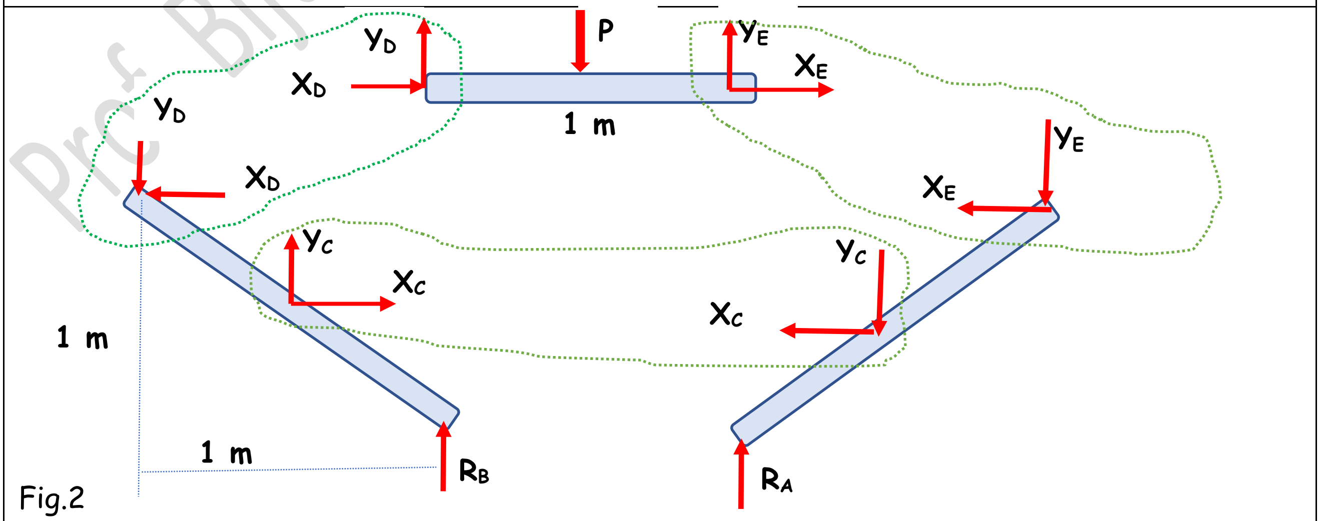


Fig.2

There are totally eight unknown forces in the free body diagrams Fig 2. We can write three equilibrium equations for each of the member. So, we get nine equations in total. We can solve them simultaneously to evaluate the unknows.

To make the steps simpler, let us calculate the support reactions from the free body diagram of the whole frame.

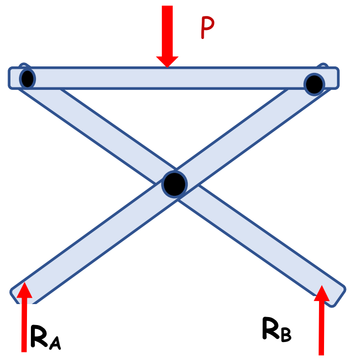


Fig. 3

The equilibrium equations are

$\Sigma F_y = 0$

$R_A + R_B - P = 0$

$\Sigma M_A = 0$

$R_B .\ 1 - P.\frac{1}{2} = 0$

$R_B = 0.5P$

$R_A = 0.5P$

Now con consider member DE

$\Sigma F_x = 0$

$X_D + X_E = 0$

$\Sigma F_y = 0$

$Y_D + Y_E - P = 0$

$\Sigma M_D = 0$

$Y_E.1 - P.\frac{1}{2} = 0$

$Y_E = 0.5P$

$Y_D = 0.5P$

(Both the values are +ve. Hence the directions of Y_E and Y_D shown on member DE are correct)

Free body diagram of member BD

$\Sigma F_y = 0$

$R_B + Y_c - Y_D = 0$

$0.5P + Y_C - 0.5P = 0$

$Y_C = 0$

$\Sigma M_c = 0$

$R_B \times 0.5 + Y_D \times 0.5 + X_D \times 0.5 = 0$

$0.5P \times 0.5 + 0.5P \times 0.5 + X_D \times 0.5 = 0$

$X_D = -P$

$X_E = P$

$\Sigma F_x = 0$

$X_C - X_D = 0$

$X_C = -P$

(Both the values of X_D and X_C are -ve. Hence the directions of X_D and X_C shown on member AD should have been opposite)

We have obtained all the unknown forces. Hence, there is no need of conisdering the free body diagram of BE. This is because we have found out the support reactions with the help of the free body diagram of the whole frame.

Q.2 Another frame formed by three members is shown in figure 2. A load of 12 kN is acting at D.The pin at C can slide along the slot without friction. Both the supports are hinged at E and F. TTDetermine the forces in the pins at A, B, C, E, and F.

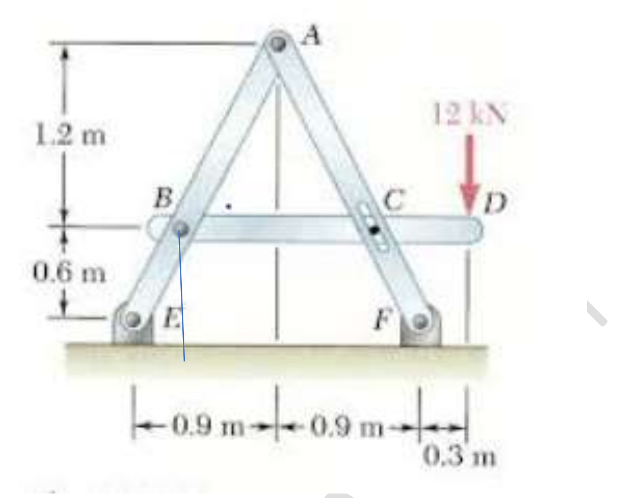


Fig.4

Let us start by drawing the free body diagrams of each member

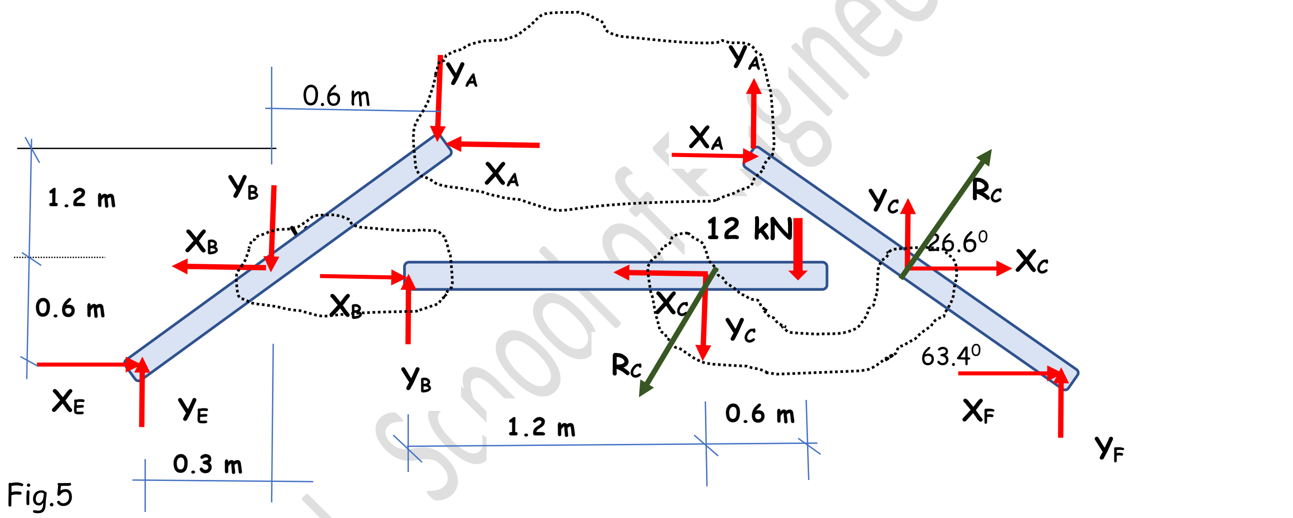


Fig.5

The total number of unknown forces in the three free body diagrams are 10. But we can have only a maximum of 9 equations from the three diagrams. Since it is given that the pin at C can slide freely along the slot, the force at C will be always normal to the slot (no friction). The bar AF is inclined at an angle $\theta = \tan^{-1}(1.8/0.9) = 63.4^0$ with horizontal. Therefore, R_C which is perpendicular toAF is inclined at 26.6^0 with horizontal.

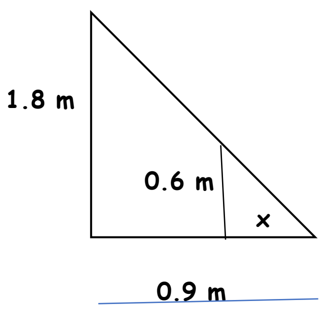
$X_C = R_C \cos 26.6$ and $Y_C = R_C \sin 26.6$

Now we have only 9 unknowns, which can be solved for.

We need to calculate the distance BC. For that consider the similar triangles shown

$$\frac{x}{0.9} = \frac{0.6}{1.8} \quad \gg \boldsymbol{x = 0.3\,m}$$

Hence BC=1.8-2x =1.2 m and hence BD=1.2+0.0.3+0.3=1.8m



Consider the free body diagram of the whole frame.

$\Sigma F_x = 0$

$X_F + X_E = 0$

$\Sigma F_y = 0$

$Y_F + Y_E - 12\text{kN} = 0$

$\Sigma M_E = 0$

$Y_F \times 1.8\text{m} - 12\text{ kN} \times 2.1\text{m} = 0$

$Y_F = 14\text{ kN}$

$Y_E = -2\text{ kN}$

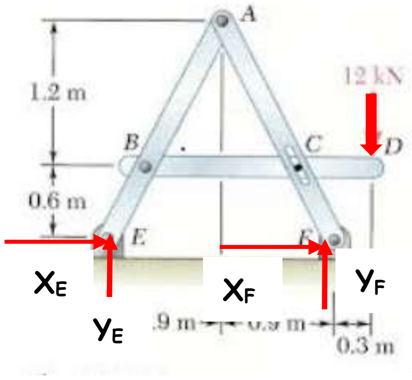


Fig. 6

Consider the free body diagram of member BD

$\Sigma M_B = 0$

$-Y_C \times 1.2\text{m} - 12\text{ kN} \times 1.8\text{m} = 0$

$Y_C = -18\text{ kN}$

$R_C = -40\text{ kN}$

$X_C = -35.8\text{ kN}$

$\Sigma F_y = 0$

$Y_B - Y_C - 12\text{kN} = 0$

$Y_B = -6\text{ kN}$

$\Sigma F_x = 0$

$X_B + X_C = 0$

$X_B = 35.8\text{ kN}$

(The values of Y_C , Y_B and X_C are -ve. Hence the directions of Y_C , Y_B and X_C shown on member BD should be opposite)

Free body diagram of member AF

$\Sigma F_y = 0$

$Y_A + Y_C + Y_F = 0$

$Y_A - 18 + 14 = 0$

$Y_A = 4\text{ kN}$

$\Sigma M_A = 0$

$Y_C \times 0.6 + X_C \times 1.2\text{m} + Y_F \times 0.9 + X_F\, 1.8 = 0$

$-18 \times 0.6 - 35.8 \times 1.2 + 14 \times 0.9 + X_F \times 1.8 = 0$

$X_F = 20.8\text{ kN}$

$X_E = -20.8\text{ kN}$

(The value of X_E is -ve. Hence the direction of X_E shown on the FBD of the frame should be opposite)

$\Sigma F_x = 0$

$X_A + X_F + X_C = 0$

$X_A = -56.6\text{ kN}$

The value of X_A are -ve. Hence the direction of X_A shown on member AF should be opposite)

Q3. Find the forces in the pins at A, B, C, D, and E, if a load P=1 kN acts at A.

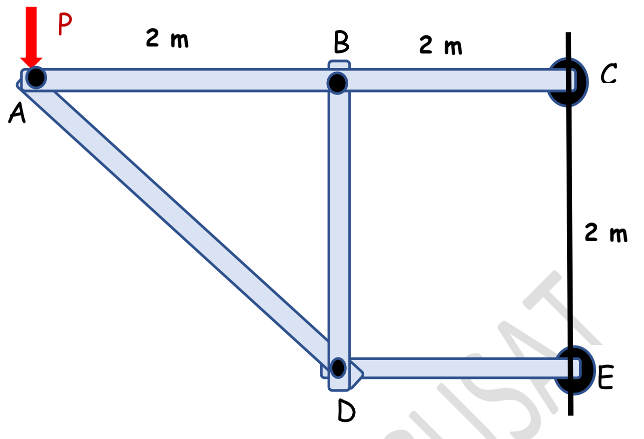


Fig. 7

It can be see that the members AD, DE, and BD are having connections only at their ends and no forces are acting at places other than the joints. So they are 2-force members. For equilibrium of these members,the forces at their ends should be equal, opposite, and collinear and they must act along the bars. Bar AC is having forces at points A, B, and C. The inclination of bar AD is 45^0

Let us draw the free body diagram of the whole frame first.

$\Sigma M_E = 0$

1 kN x 4m-X_C x 2m =0

X_C = 2kN

$\Sigma F_X = 0$

$X_C + X_E$ =0

X_E = -2kN

$\Sigma F_y = 0$

$Y_C + Y_E - 1$ =0

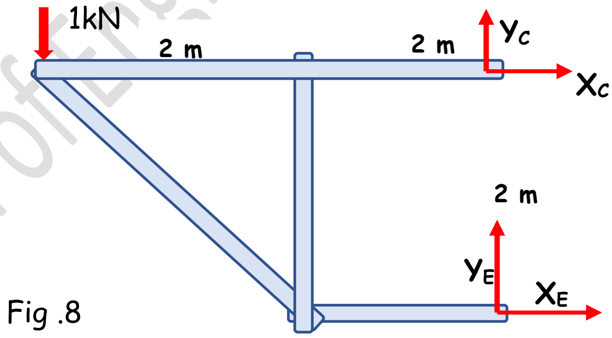


Fig .8

Consider the free body diagrams of the individual members. The directions of the forces in the 2-force members are arbitrarily taken as tensile forces (away from the joint).

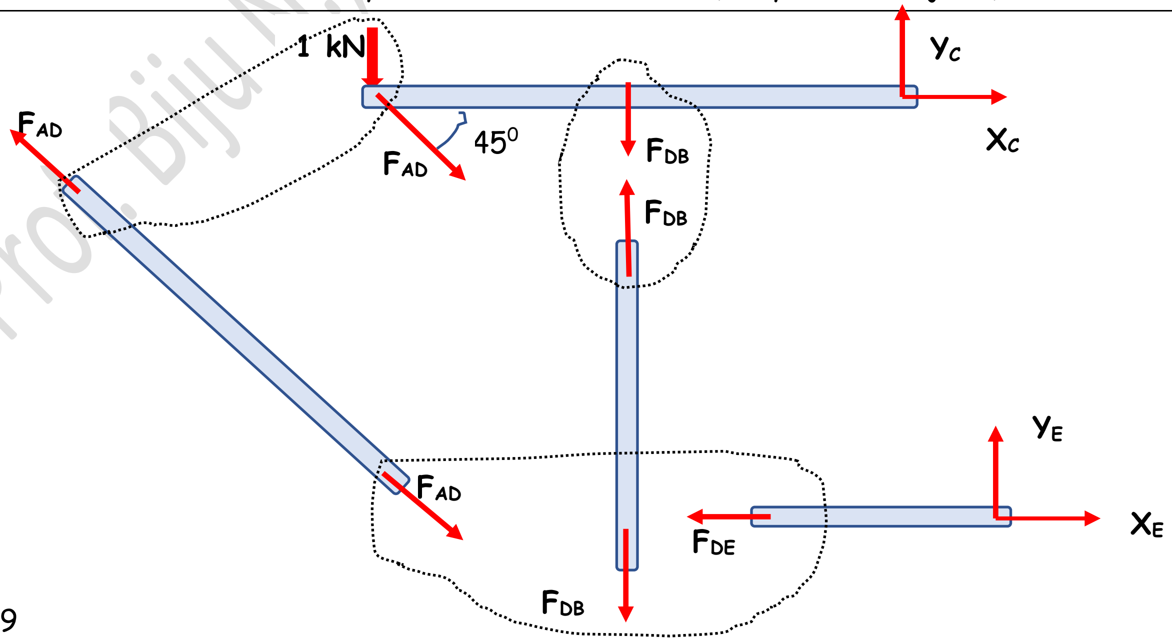


Fig.9

Looking at the free body diagram of member DE, for its equilibrium the force at D and E should be equal, and opposite and collinear. Therefore, the component $Y_E = 0$.

$F_{DE} = X_E = -2$ kN

We can get the same result if we consider $\Sigma M_D = 0$, which will give $Y_E \times 2m = 0$

»» $Y_E = 0$

$Y_C = 1$ kN (from the FBD of the whole frame)

Free body diagram of the member AC

$\Sigma M_A = 0$

$Y_C \times 4m - F_{DB} \times 2m = 0$

$F_{DB} = 2kN$

$\Sigma F_y = 0$

$Y_C - F_{DB} - F_{AD} \sin 45 - 1 = 0$

$F_{AD} = -2.83$ kN

$\Sigma F_X = 0$

$X_C + X_E = 0$

$X_C = -2kN$

The directions of F_{AD} and F_{DE}, and X_E are opposite to that we have assumed. Hence, member DE and AD are in compression while DB is in tension

#Exercise 1

Calculate the forces in the pins A,B, C, D, E, and F of the frame loaded and supported as shown.

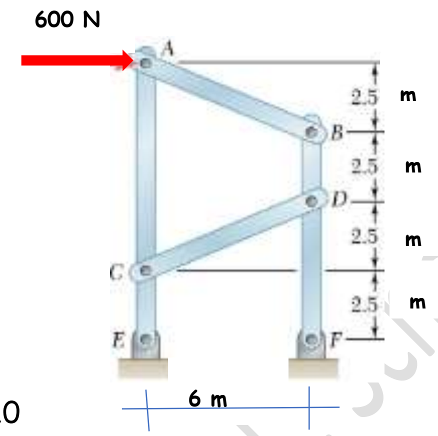
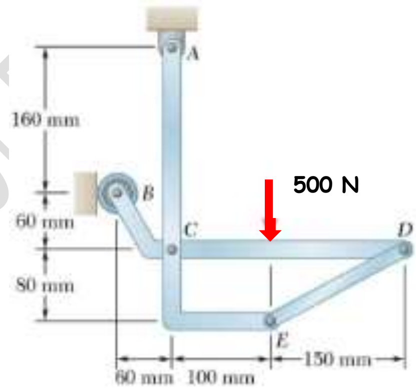


Fig. 10

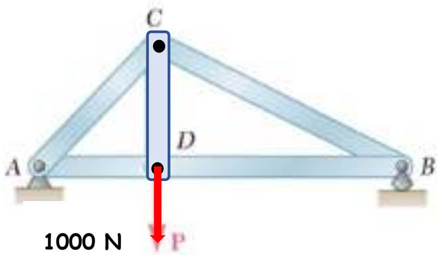
#Exercise 2

Calculate the forces in the pins A,B, C, D, and E of the frame loaded and supported as shown.



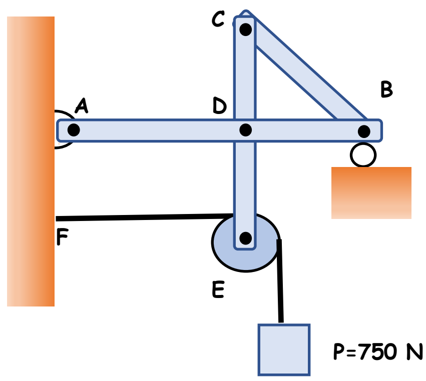
#Exercise 3

Calculate the forces in the pins A,B, C, and D of the frame loaded and supported as shown. Take AB=3m, AD=1m, CD=1



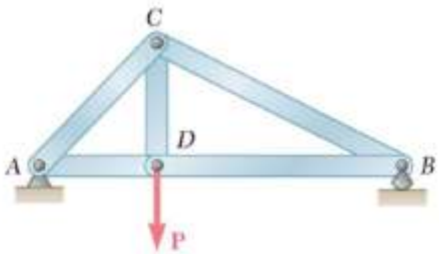
#Exercise 4

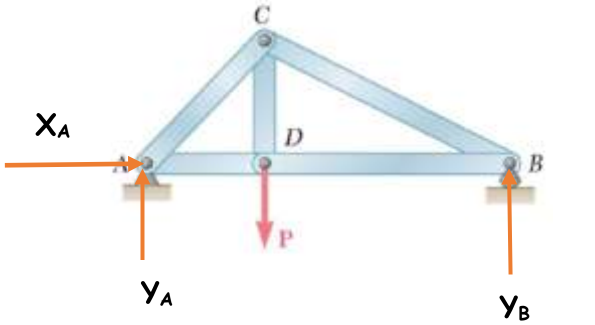
Calculate the forces in the pins A,B, C, and D of the frame loaded and supported as shown. Assume AD=1.2m, DB=1.2m, CD= 1m, DE= 1m.



#Exercise 3

Calculate the forces in the pins A,B, C, and D of the frame loaded and supported as shown. Take AB=3m, AD=1m, CD=1m





$\Sigma F_X = 0$

$X_A = 0$

$\Sigma M_A = 0$

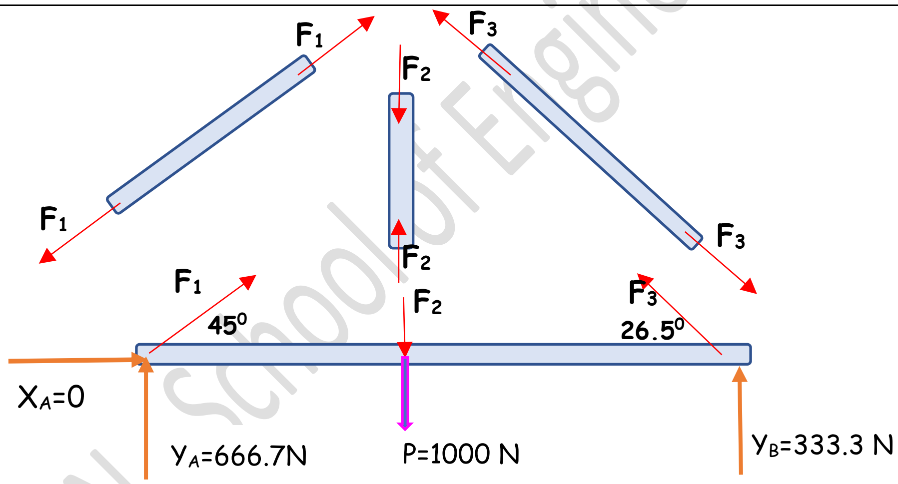
$-1000\text{ N} \times 1\text{m} + Y_B \times 3\text{m} = 0$

$Y_B = 333.3\text{ N}$

$\Sigma F_y = 0$

$Y_B + Y_A - 1000 = 0$

$Y_A = 666.7\text{ N}$



$\Sigma F_X = 0$

$X_A + F_1 \cos 45 - F_3 \cos 26.5 = 0$

$F_1 = 1.26 F_3$

$\Sigma F_y = 0$

$Y_A + Y_B + F_1 \sin 45 + F_3 \sin 26.5 - F_2 = 0$

$666.7 + 333.3 - 1000 + 1.26 F_3 \sin 45 + F_3 \sin 26.5 = F_2$

$F_2 = 1.34\ F_3$

$\Sigma M_A = 0$

$-1000\text{ N} \times 1\text{m} - F_2 \times 1\text{m} + Y_B \times 3\text{m} + F_3 \sin 26.5 \times 3\text{m} = 0$

$-1000 - 1.34\ F_3 + 1000 + F_3 \sin 26.5 \times 3\text{m} = 0$

$F_3 = 0$

$F_1 = F_2 = 0$